

YouTube Trending Video Analytics

A Data Storytelling Report

Overview

Every day, YouTube shows a list of trending videos to millions of people. This project looks at what makes a video trend, and whether trending videos are mostly local to one country or popular everywhere. The dataset has 178,580 trending video records from five countries (United States, United Kingdom, India, Japan, and Germany), collected between November 2017 and June 2018. This report covers the full process: cleaning the data, writing SQL queries, running sentiment analysis on titles, building visualizations, and putting it all together in a dashboard. The sections below explain what was done at each step and what it showed.

Process

1. Data Collection and Cleaning

The raw data came as five CSV files, one for each country, taken from a YouTube trending video dataset on Kaggle. Each file was loaded into a Jupyter notebook with Pandas, tagged with its country, and combined into one table. Date formats were cleaned up, and the merged dataset was exported to SQLite so it could be queried with SQL as well as analyzed in Python.

2. SQL Analysis

Ten SQL queries were written against the combined dataset using DB Browser for SQLite. These covered view and engagement rankings, category trends by country, like-to-dislike ratios, videos that trended in more than one country, and trending activity by month. Most of the findings in this report come directly from these queries.

3. Sentiment Analysis

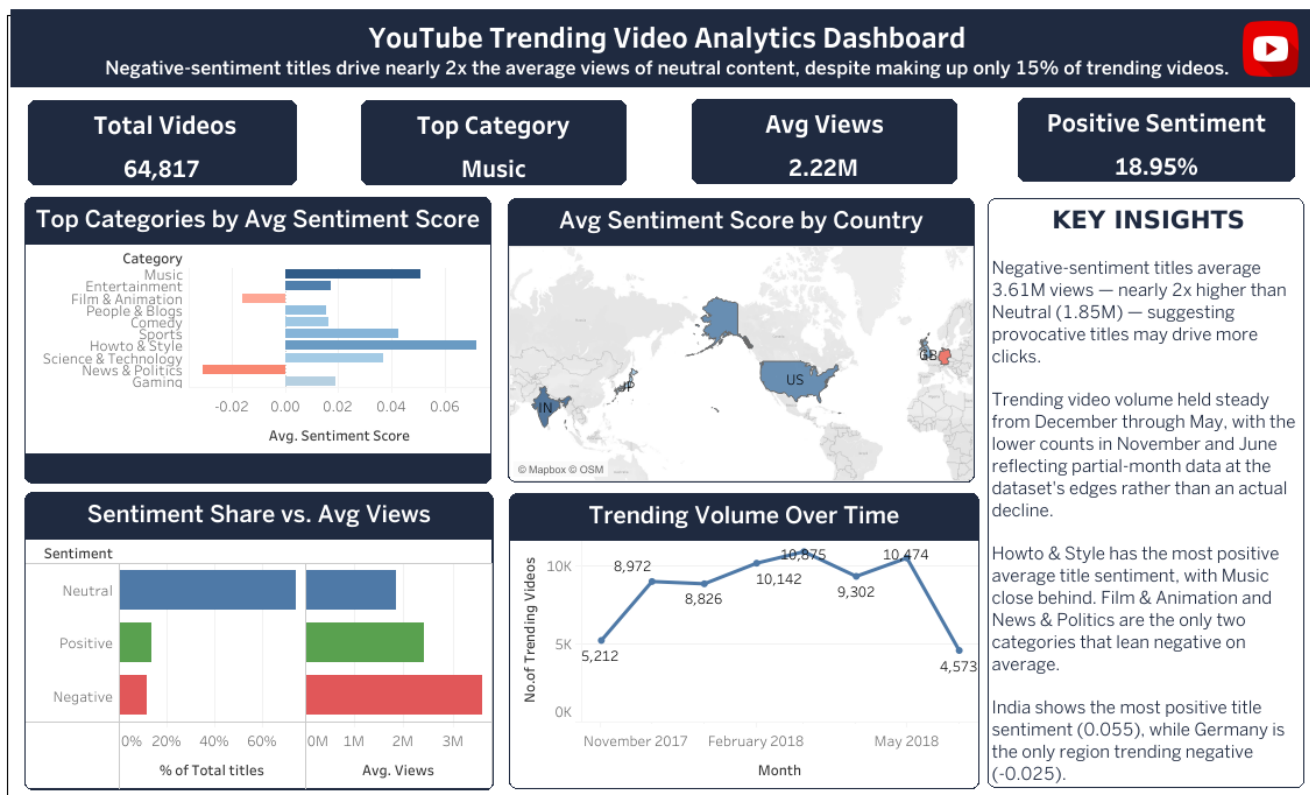
Every video title was scored using VADER (Valence Aware Dictionary and sEntiment Reasoner) and labeled Positive, Neutral, or Negative based on its sentiment score. This made it possible to check whether the tone of a title is related to how well a video performs, which view counts alone can't tell you.

4. Exploratory Visualization

Using the SQL results and sentiment scores, several visualizations were built in Python with Matplotlib and Seaborn to look at the data from different angles: categories, countries, sentiment, engagement, and trending activity over time. The most useful ones are shown in the Findings section below.

5. Dashboard

The key findings were brought together into one interactive Tableau dashboard, using a YouTube-style red, white, and navy theme. It was later updated to add a region-wise map showing average sentiment score by country, alongside the category, sentiment, and trending volume charts already there. The dashboard is published on Tableau Public so it can be explored directly, not just read about.



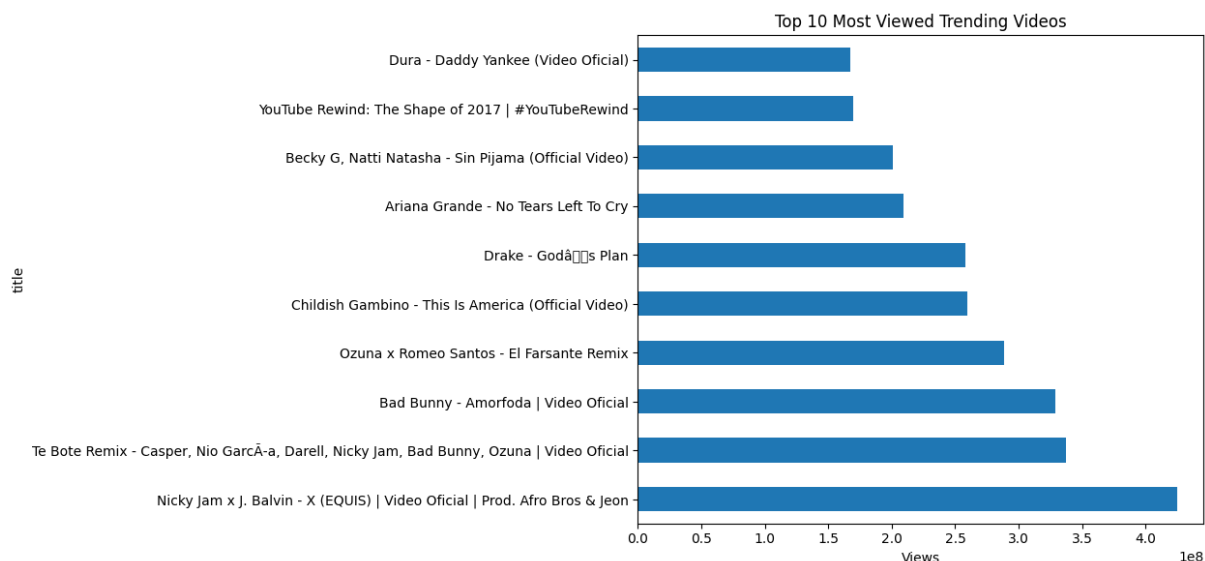
The published Tableau dashboard, combining category, region, sentiment, and time-based views.

[View the live dashboard on Tableau Public](#)

Findings

Finding 1: A Small Set of Global Hits Dominates the Top of the Views List

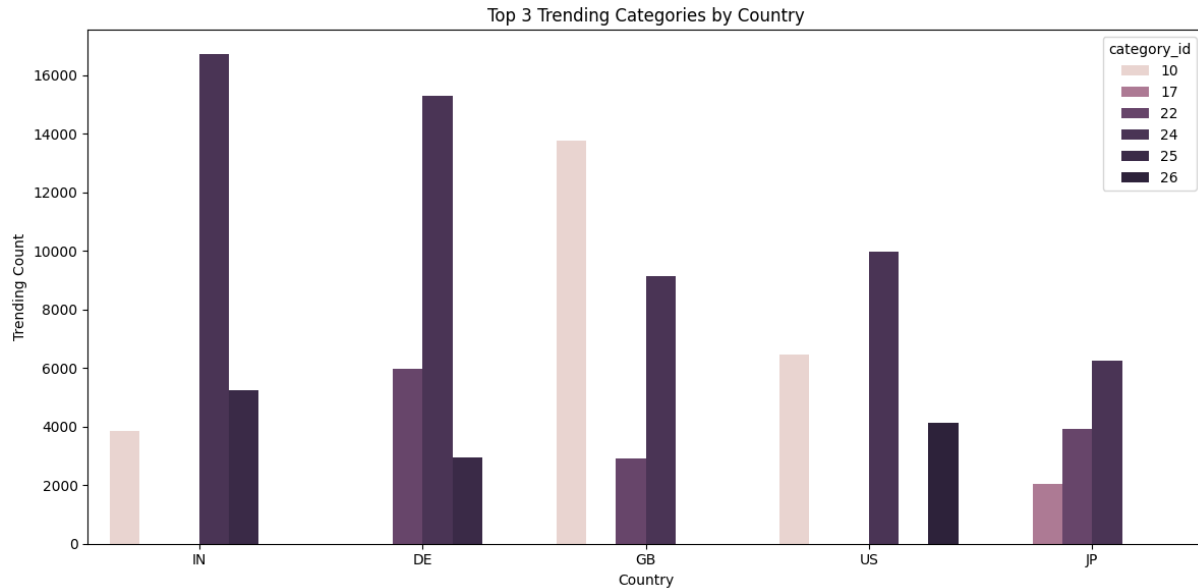
When all videos are ranked by total views, the top of the list is mostly music videos, plus one major recap special. Nicky Jam x J. Balvin's "X (EQUIS)" is at the top with over 420 million views, followed by collaborations from Bad Bunny, Ozuna, Childish Gambino, Drake, and Ariana Grande, along with YouTube's own annual Rewind video. Every single video in the top 10 is either a global music release or a YouTube institution — none of them are regional or niche videos.



Top 10 most viewed trending videos across the full dataset.

Finding 2: Category Dominance Differs by Country

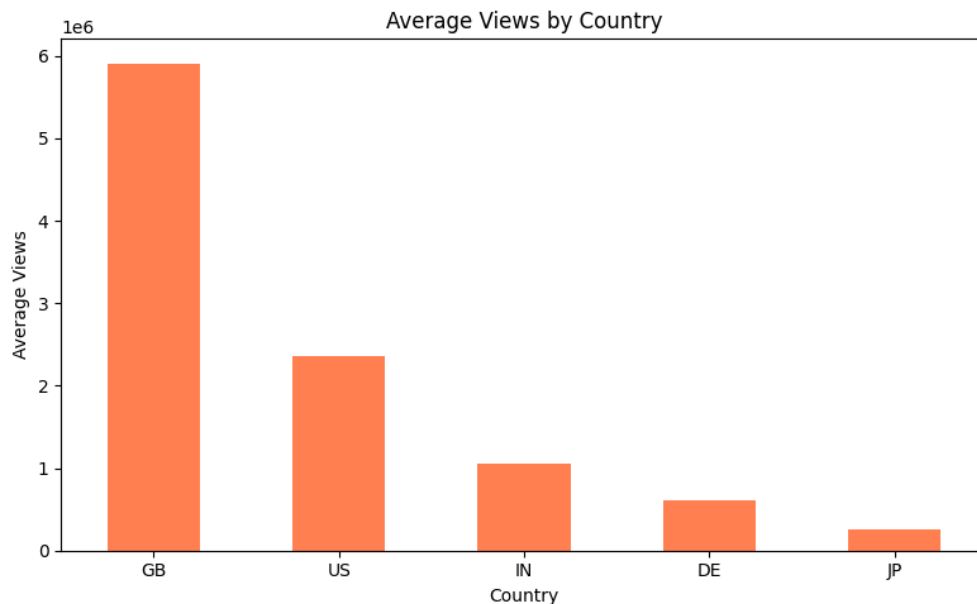
Looking at the top 3 trending categories in each country, Category 24 (Entertainment) comes first in Germany, the UK, the US, and Japan. India is the outlier here: Category 10 (Music) is by far its most frequent trending category, ahead of Entertainment and Comedy (Category 23). This kind of country-level difference would not show up if the dataset were looked at as a whole.



Top 3 trending categories by country, by number of appearances on the trending list.

Finding 3: The UK Leads on Average Views by a Wide Margin

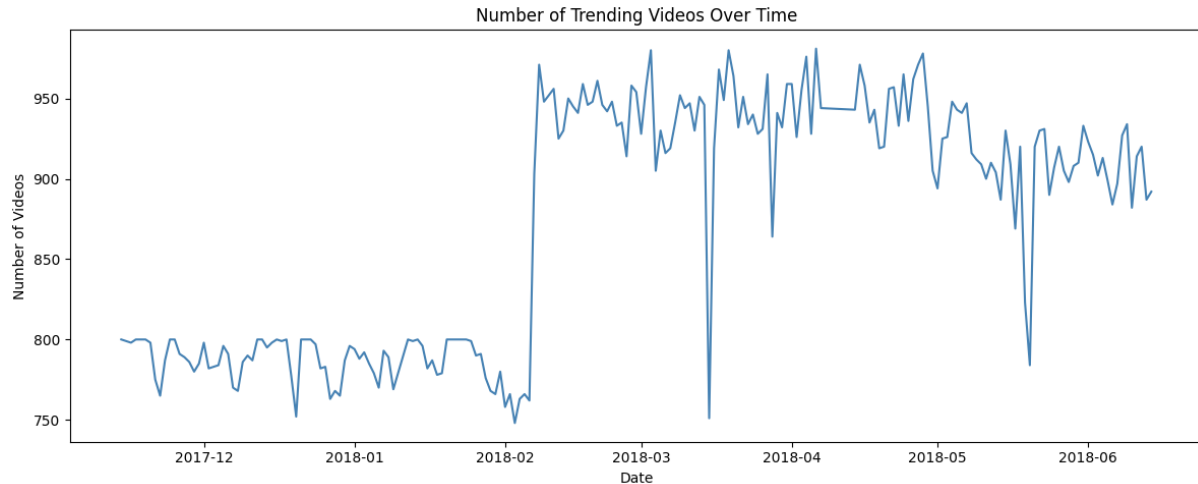
The United Kingdom has the highest average views per trending video at 5.91 million, more than double the next country, the United States, at 2.36 million. India, Germany, and Japan are all lower than that, with Japan the lowest at around 262,000 average views. The gap between countries is big enough that "trending" seems to mean a different scale of popularity depending on where you are.



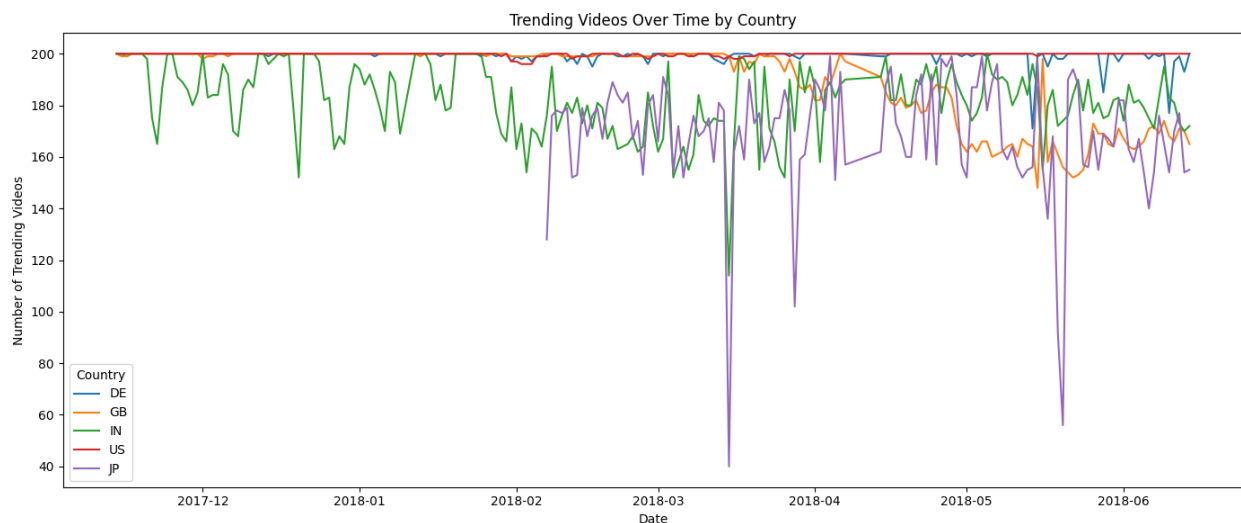
Average views per trending video by country.

Finding 4: Trending Volume Shows a Structural Shift Partway Through the Dataset

The number of trending videos per day stays fairly steady at around 800 from November 2017 to early February 2018, then jumps to around 950 per day for the rest of the dataset. This is a sudden jump, not a gradual one, which suggests the number of videos being tracked per day changed at that point rather than there being an actual surge in trending activity. Looking at it by country, the United States stays at exactly 200 trending videos per day for the whole period, while Germany, the UK, India, and Japan vary more. This is probably about how consistently each country's trending list was collected in the source data, not a real difference in behavior between countries.

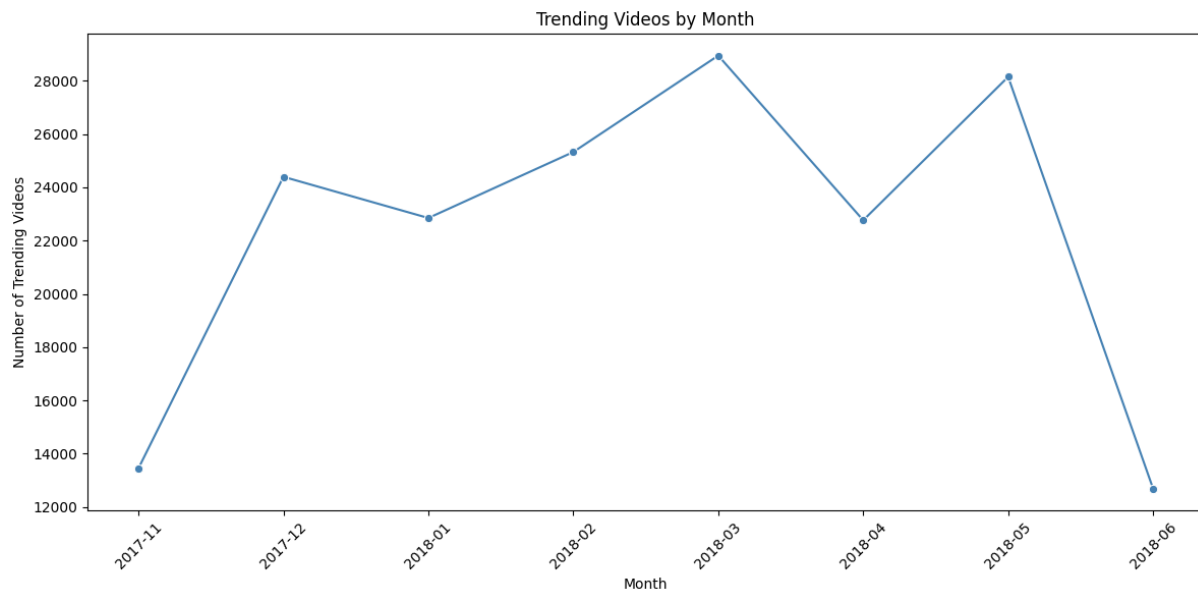


Number of trending videos per day across the full dataset window.



Daily trending video counts by country, showing the US's fixed daily count alongside more variable counts elsewhere.

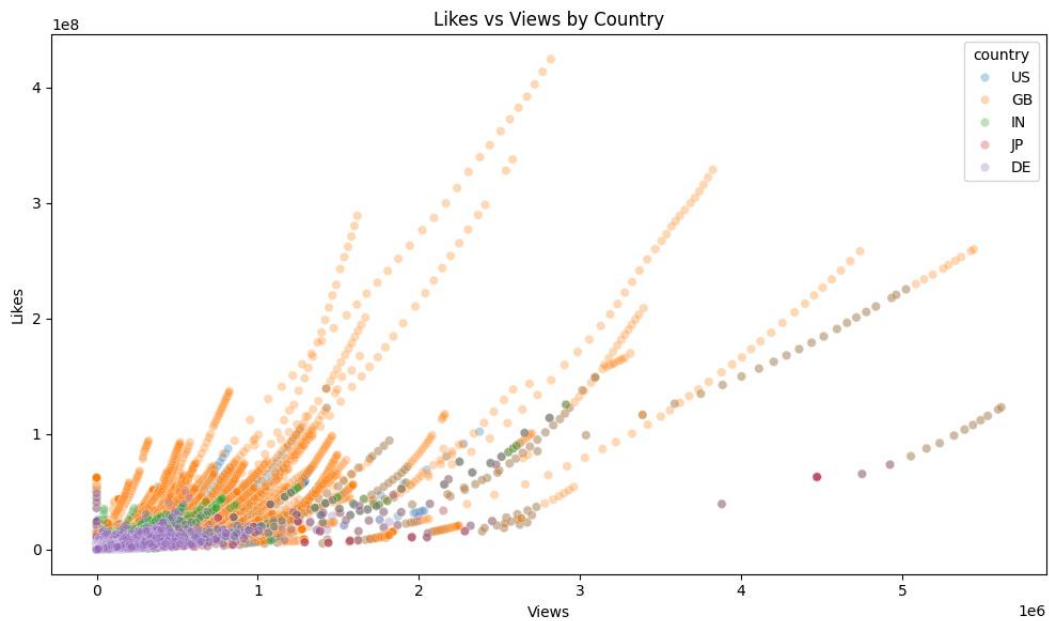
The same pattern shows up when the data is grouped by month: November 2017 and June 2018 are both partial months in this dataset and have the lowest counts, while March and May 2018 are the highest. December is often expected to be a peak month because of holiday content, but here it actually falls in the middle once the partial months at each end are accounted for.



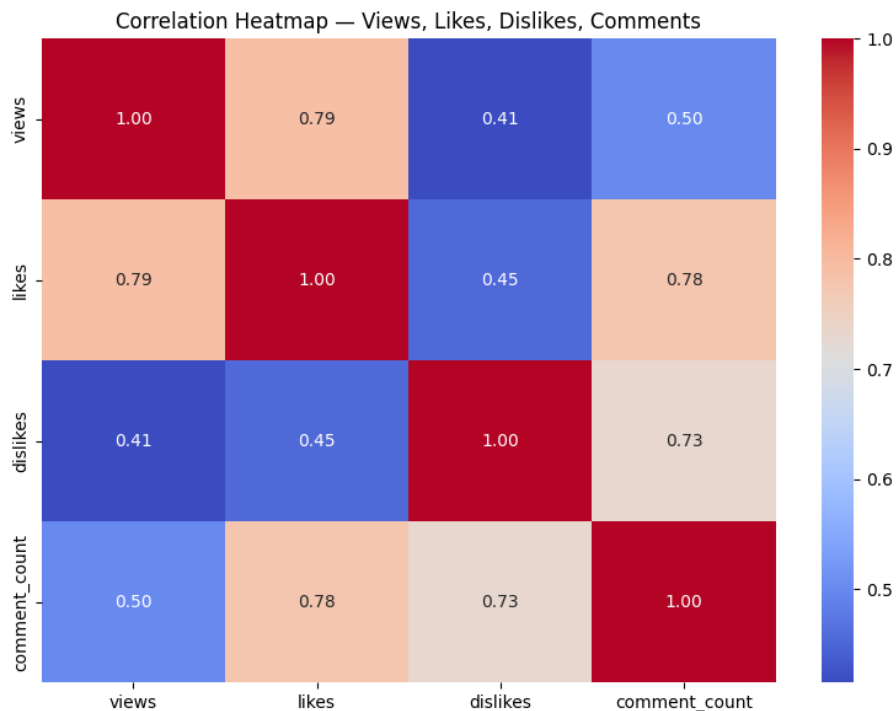
Total trending videos by calendar month.

Finding 5: Likes and Views Are Strongly Correlated, but Dislikes and Comments Less So

Likes and views move together closely across every country, with a correlation of 0.79, and this relationship is steepest for UK videos, which fits with the UK's higher average view counts. Comment count is moderately correlated with both likes (0.78) and dislikes (0.73), but dislikes only correlate weakly with views (0.41). This means a high view count doesn't reliably tell you how much negative reaction a video got.



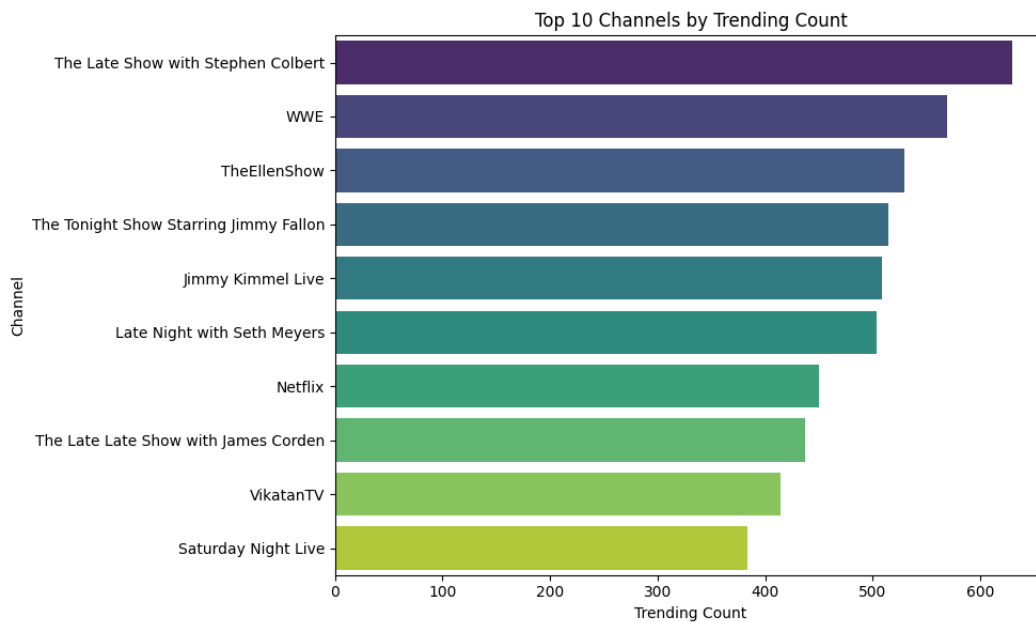
Likes vs. views by country, showing the UK's steeper view-to-like relationship.



Correlation heatmap across views, likes, dislikes, and comment count.

Finding 6: Talk Shows and Recurring Channels Dominate Trending Frequency

Looking at how often channels appear on the trending list, the top spots are almost all daily-upload talk shows and recurring shows: The Late Show with Stephen Colbert (630 appearances), WWE (569), TheEllenShow (529), The Tonight Show Starring Jimmy Fallon (515), and Jimmy Kimmel Live (509). This is a different list from the most-viewed videos — these channels trend often simply because they upload all the time, not because any one video goes viral.

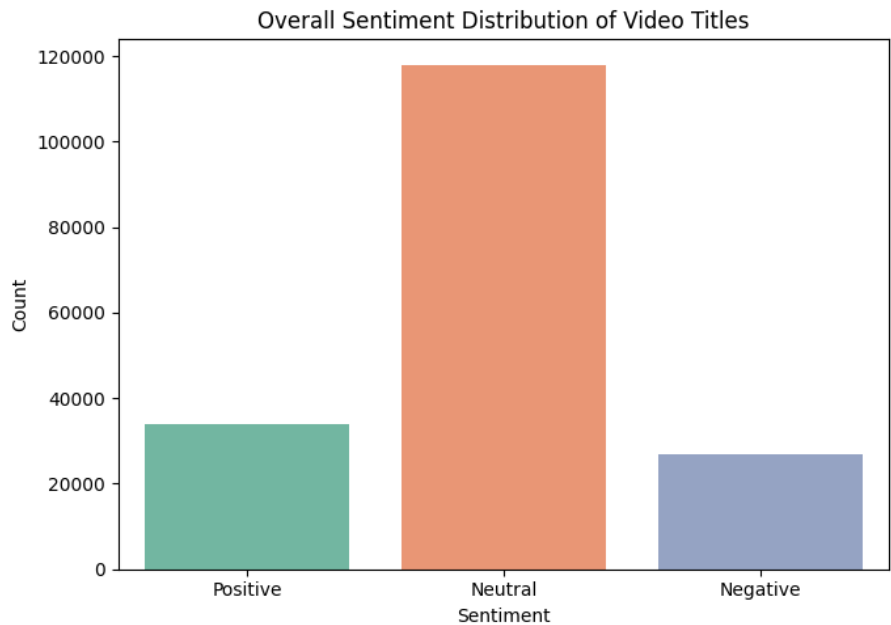


Top 10 channels ranked by number of trending appearances.

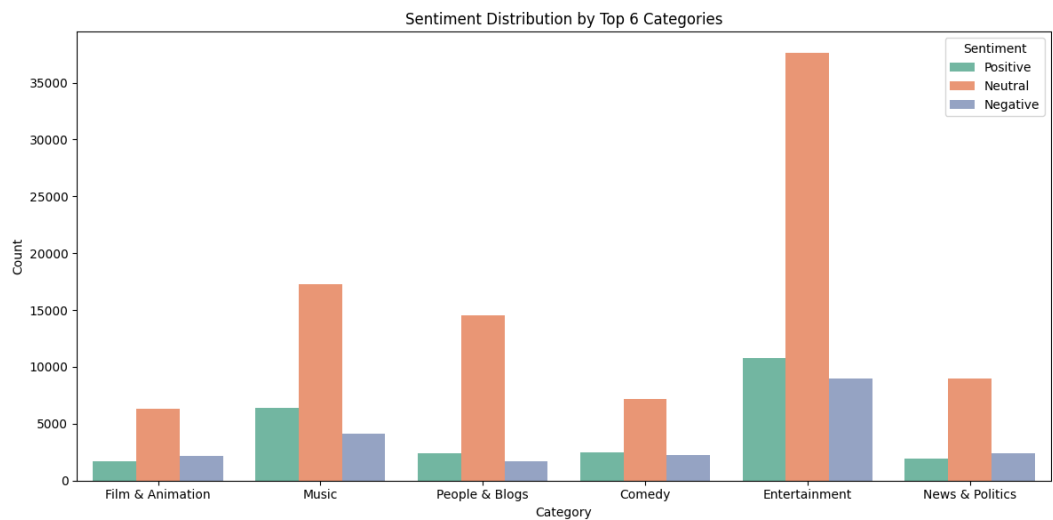
Finding 7: Negative-Sentiment Titles Draw Nearly Double the Views

Across the full dataset, most video titles are scored Neutral (about 118,000 out of 178,580), with around 34,000 Positive titles and only around 27,000 Negative titles. So negative titles are clearly the minority,

not the norm. But negative-sentiment videos still average 3.61 million views, compared to 1.85 million for neutral titles and 2.42 million for positive ones. So the least common sentiment category actually gets close to double the average views of the most common one.

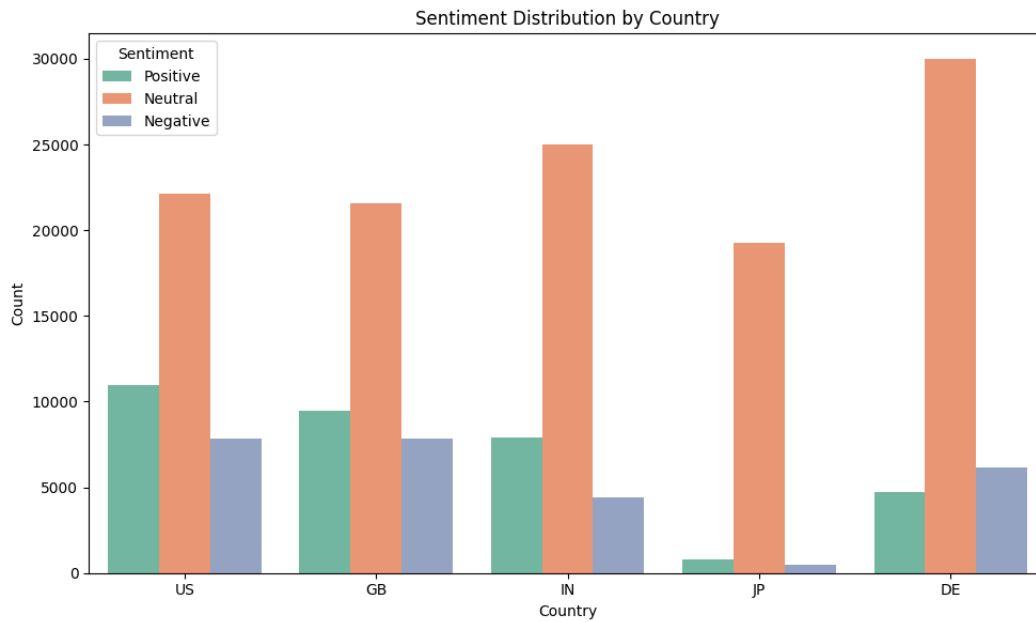


Overall sentiment distribution of video titles across the dataset.

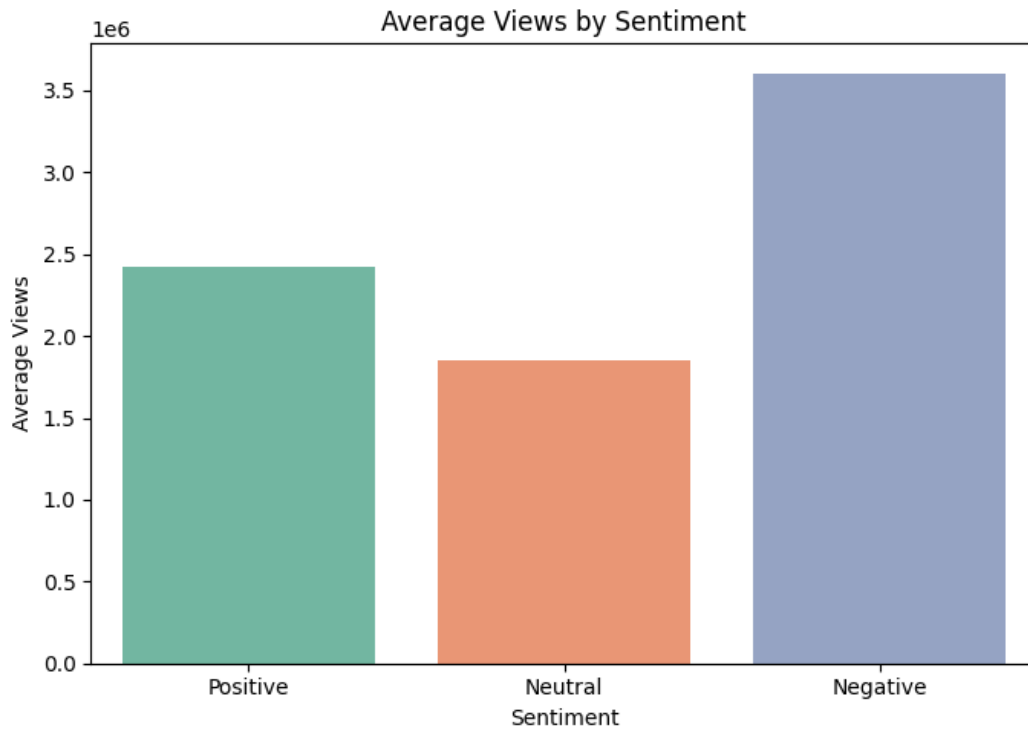


Sentiment distribution across the top 6 trending categories.

This same pattern shows up at the country level. Neutral is the largest sentiment group everywhere, but Japan has an especially small Positive and Negative share compared to its Neutral majority, and Germany has the highest Negative-sentiment count of any country. Looking at average sentiment score instead of sentiment share, India has the most positive average title sentiment of the five countries, followed by the US and UK. Germany is the only country with a negative average sentiment score, which fits with it having the most negative-leaning titles overall. By category, Entertainment has the most videos across all three sentiment types, and Music has a noticeably higher share of Positive titles compared to Negative ones than most other categories do — which makes sense given how upbeat most music release titles tend to be.



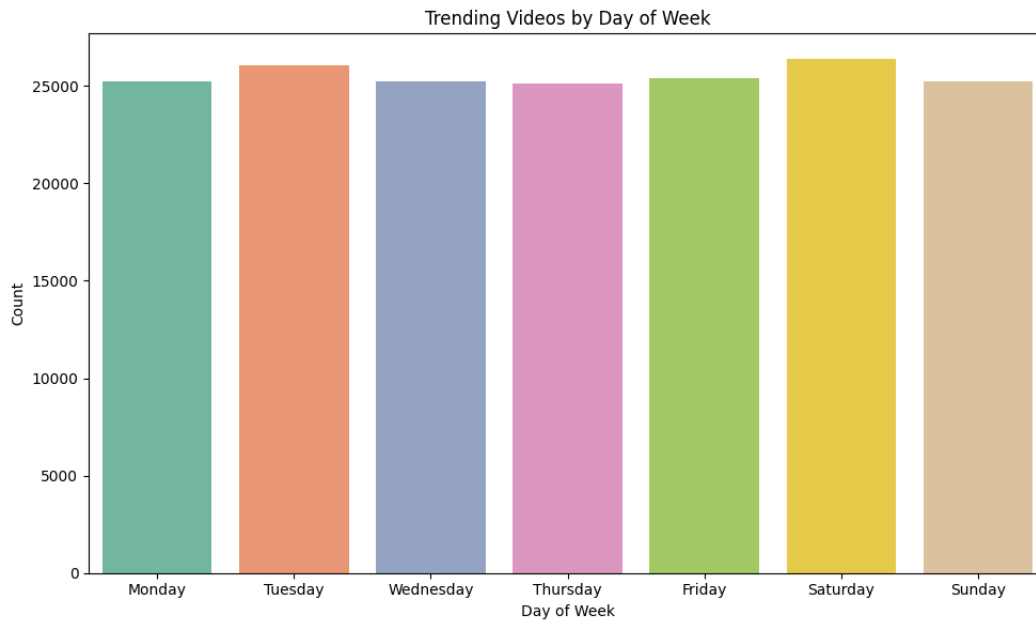
Sentiment distribution by country.



Average views by sentiment category.

Finding 8: Trending Activity Is Fairly Even Across Days of the Week

Trending video counts by day of the week are pretty flat. Saturday and Tuesday are a little higher than the other days, but the gap between the highest and lowest day is small compared to the overall volume. This suggests trending activity in this dataset isn't strongly tied to a weekly upload or viewing pattern.



Trending video counts by day of the week.

What This Means

Putting these findings together, trending performance on YouTube seems to come from a few different factors, not just one. Title sentiment does have a measurable link to view count, even though negative titles are a minority of what gets posted. Real cross-border virality is rare and mostly limited to major studio and label releases — most trending videos stay local to their own country. Country matters just as much as the content itself: the UK's trending list looks nothing like Japan's or Germany's, both in scale and in what seems to drive engagement. And there's a separate path to trending that has nothing to do with going viral — daily talk shows and recurring channels trend constantly just by uploading consistently, not by having any one breakout video.

Limitations

- The dataset only covers a seven-month window (November 2017 to June 2018), so these findings reflect that period and might not apply to how YouTube trends or its algorithm work now.
- The jump in daily trending volume around February 2018, and the US's perfectly flat daily count, are both more likely explained by how consistently the data was collected across countries and time, not by any real-world event.
- Sentiment was scored using VADER, mainly on video titles, which is what the dashboard and findings above are based on. Tags were also scored separately as a check, but full descriptions and thumbnails weren't analyzed, even though they probably influence whether someone clicks too.
- The link between sentiment and views is just a correlation. This analysis doesn't prove that negative titles cause higher views — only that the two show up together in this dataset.